Engineer to Excel

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DECISION TREE LEARNING AND REPRESENTATION

ITA 0608 - MACHINE LEARNING

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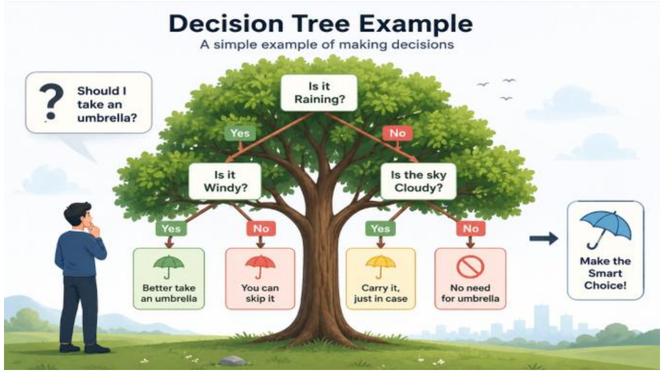
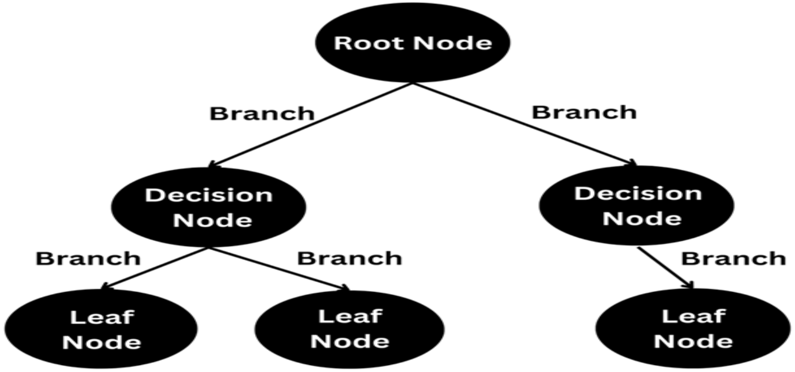
What is a Decision Tree?

- Supervised learning algorithm
- Used for classification and regression
- Tree-like decision model
- Easy to understand and visualize



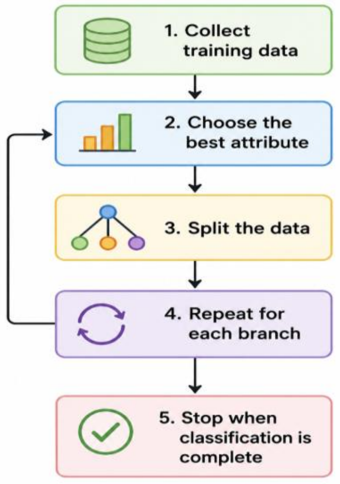
STRUCTURE

- **Root Node** – Starting point
- **Decision Node** – Condition/test
- **Branch** – Test outcome
- **Leaf Node** – Final decision



WORKING

1. Collect training data
2. Choose the best attribute
3. Split the data
4. Repeat for each branch
5. Stop when classification is complete



ENTROPY

- Measures data impurity
- Lower entropy → Better classification
- Higher entropy → More uncertainty

Formula:

$$Entropy(S) = -\sum p_i \log_2(p_i)$$

INFORMATION GAIN

Information Gain

Information Gain measures how much an attribute reduces uncertainty.

Formula

$$IG = Entropy(Parent) - Weighted\ Entropy(Children)$$

Key Points

- ➢ Higher Information Gain → Better attribute.
- ➢ The attribute with the highest gain becomes the next decision node.
- ➢ Used by the ID3 Decision Tree Algorithm.

TREE PRUNING

What is Tree Pruning?

Pruning removes unnecessary branches from the tree to improve performance.

Benefits

- Reduces overfitting.
- Improves generalization.
- Simplifies the model.

Types

1. Pre-Pruning
2. Post-Pruning

ADVANTAGES AND LIMITATIONS

Advantages

1. Easy to understand and interpret.
2. Simple visualization.
3. Handles numerical and categorical data.
4. Requires little data preprocessing.
5. Fast prediction.

Limitations

- Can overfit the training data.
- Sensitive to noisy data.
- Small data changes may create different trees.
- Deep trees become complex.
- Lower accuracy than ensemble methods in some cases.

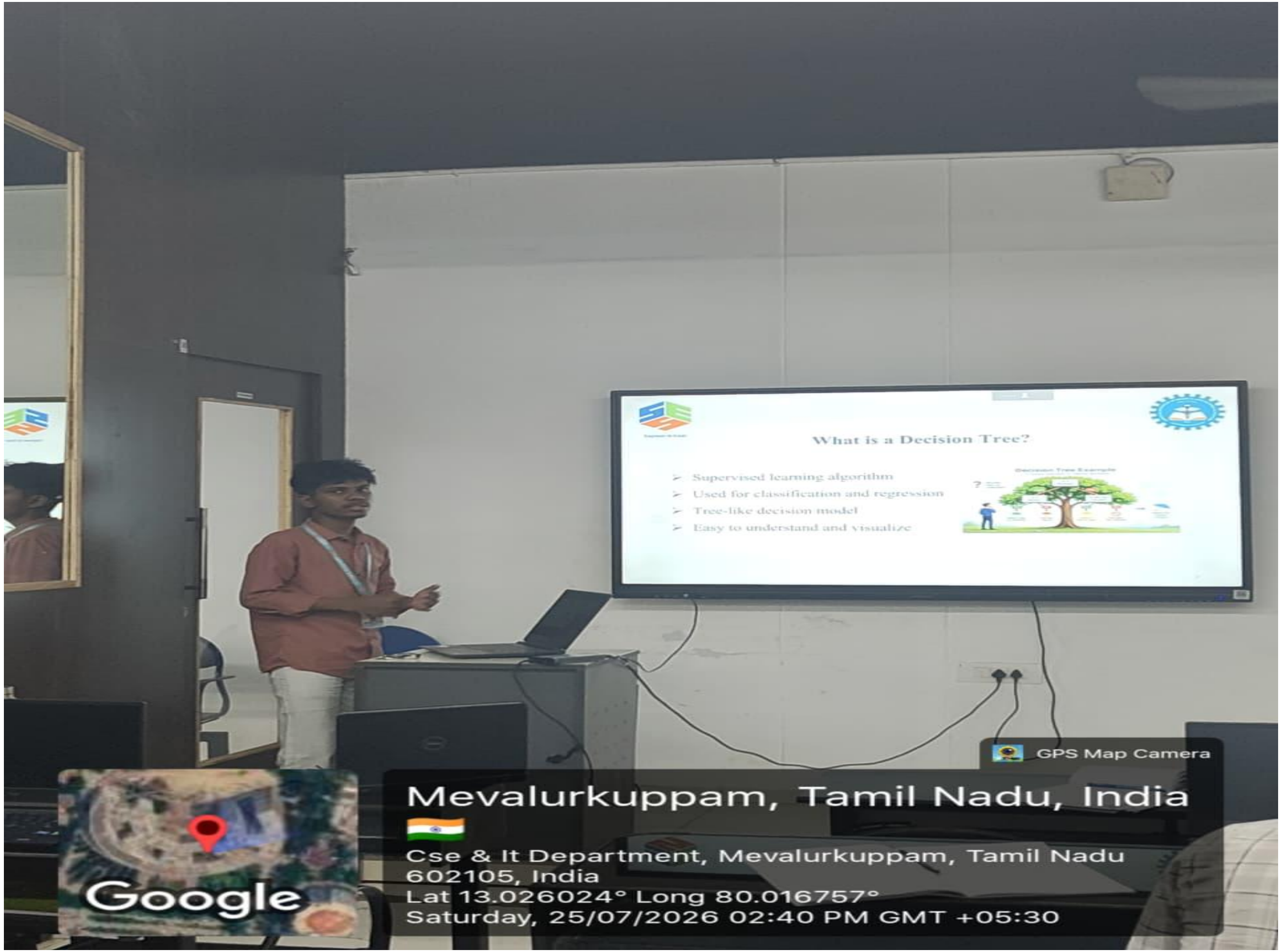
Applications of Decision Trees

1. Medical diagnosis
2. Credit risk analysis
3. Fraud detection
4. Customer segmentation
5. Loan approval systems
6. Spam email detection
7. Product recommendation
8. Business decision support

CONCLUSION

Decision Tree is a simple and effective machine learning algorithm used for classification and decision-making. It builds a tree structure using entropy and information gain, while pruning helps improve accuracy by reducing overfitting. Its ease of understanding and implementation makes it widely used in many real-world applications.

Thank You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